

2.02 Data Presentation and Interpretation				
2.02a	Single variable data	a) Be able to interpret tables and diagrams for single-variable data. <i>e.g. vertical line charts, dot plots, bar charts, stem-and-leaf diagrams, box-and-whisker plots, cumulative frequency diagrams and histograms (with either equal or unequal class intervals). Includes non-standard representations.</i>		ML1
2.02b		b) Understand that area in a histogram represents frequency. <i>Includes the link between histograms and probability distributions.</i> <i>Includes understanding, in context, the advantages and disadvantages of different statistical diagrams.</i>		
2.02c	Bivariate data	c) Be able to interpret scatter diagrams and regression lines for bivariate data, including recognition of scatter diagrams which include distinct sections of the population. <i>Learners may be asked to add to diagrams in order to interpret data, but not to draw complete scatter diagrams.</i> <i>[Calculation of equations of regression lines is excluded.]</i>		ML2
2.02d		d) Be able to understand informal interpretation of correlation.		
2.02e		e) Be able to understand that correlation does not imply causation.		

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
2.02f	Measures of average and spread	f) Be able to calculate and interpret measures of central tendency and variation, including mean, median, mode, percentile, quartile, inter-quartile range, standard deviation and variance. <i>Includes understanding that standard deviation is the root mean square deviation from the mean.</i> <i>Includes using the mean and standard deviation to compare distributions.</i>		ML3
2.02g	Calculations of mean and standard deviation	g) Be able to calculate mean and standard deviation from a list of data, from summary statistics or from a frequency distribution, using calculator statistical functions. <i>Includes understanding that, in the case of a grouped frequency distribution, the calculated mean and standard deviation are estimates.</i> <i>Learners should understand and be able to use the following formulae for standard deviation:</i> $\sqrt{\frac{\sum(x - \bar{x})^2}{n}} = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2},$ $\sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}} = \sqrt{\frac{\sum fx^2}{\sum f} - \bar{x}^2}$ <i>[Formal estimation of population variance from a sample is excluded. Learners should be aware that there are different naming and symbol conventions for these measures and what the symbols on their calculator represent.]</i>		ML3

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
2.02h 2.02i 2.02j	Outliers and cleaning data	h) Recognise and be able to interpret possible outliers in data sets and statistical diagrams. i) Be able to select or critique data presentation techniques in the context of a statistical problem. j) Be able to clean data, including dealing with missing data, errors and outliers. <i>Learners should be familiar with definitions of outliers:</i> <i>more than $1.5 \times$ (interquartile range) from the nearer quartile</i> <i>more than $2 \times$ (standard deviation) away from the mean.</i>		ML4